

Alan Hu . Jarvis Consulting

Recent McGill University graduate in Statistics and Computer Science with a strong foundation in data engineering, analytics, software development, and problem-solving. Certified in IBM Data Engineering and Snowflake Data Engineering, with hands-on experience building and supporting scalable end-to-end data pipelines using Databricks, PySpark, Delta Live Tables, SQL, Python, Azure, and Medallion Architecture. Highly motivated to continuously learn and adopt emerging technologies that improve data reliability, performance, and business impact. Known for strong communication skills, analytical thinking, and a collaborative mindset, with a genuine enthusiasm for working directly with stakeholders to translate complex business requirements into practical, high-quality technical solutions. Well-suited for technical consulting and data engineering environments that demand both engineering rigor and client-facing excellence.

Skills

Proficient: Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Python (Pandas, Sci-kit), ETL Pipelines

Competent: Java, Databricks, Maven, Docker, Apache Spark/PySpark, Apache Airflow

Familiar: NoSQL/MongoDB, GCP/BigQuery, AWS, Snowflake, Power BI

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AlanHu

Cluster Monitor [GitHub]: Automated Linux cluster analysis (LCA) and node orchestration using Bash scripting and PostgreSQL to centralize and analyze cluster resource metrics. Containerized data collection and reporting workflows with Docker and managed version control using Git to ensure reproducibility and collaborative development. Enabled scalable SQL-based reporting on node usage trends to support data-driven capacity planning and resource allocation.

Core Java Apps [GitHub]:

- Twitter App: Placeholder to allow profile_app.sh to work.
- JDBC App: Placeholder to allow profile_app.sh to work.
- Grep App: Built a Java-based grep application to recursively scan large directory structures, apply regex-based filtering, and persist matched results to disk with efficient I/O handling. Implemented modular, interface-driven design with robust exception handling, UTF-8 encoding, and stream-based file traversal using Java NIO. Managed dependencies and project lifecycle using Maven, and integrated structured logging with SLF4J and Log4j for observability.

Python Data Analytics [GitHub]: Spearheaded a Python-based data analysis PoC for a retail gift shop to diagnose stagnant revenue performance. Used Pandas, NumPy, Jupyter Notebook, and SQL to perform exploratory analysis, customer segmentation, and trend analysis, delivering targeted recommendations aligned with the shop's customer base to drive higher engagement and incremental sales.

Highlighted Projects

Databricks Bullhorn CRM Pipeline: Maintained and enhanced enterprise ETL pipelines in Databricks using PySpark, Delta Live Tables, and Medallion Architecture (Bronze, Silver, Gold) to ingest, transform, and curate CRM data for downstream reporting and analytics. Developed REST API-based ingestion workflows using token authentication, orchestrated tier-specific pipelines through centralized Databricks Jobs, and implemented testing pipelines and validation checks to ensure data quality, code reliability, and production readiness.

E-commerce Data Platform Architecture [GitHub]: Developed a hybrid on-premise/cloud data platform for an e-commerce company, integrating MySQL for OLTP and MongoDB for NoSQL storage. Built automated ETL pipelines using Python and Apache Airflow to load data into a PostgreSQL staging warehouse and Hadoop for large-scale analytics. Applied PySpark and SparkML to support big data processing and sales forecasting within the analytics pipeline. Enabled business intelligence reporting and KPI dashboards through IBM Cognos Analytics to support data-driven decision-making.

Professional Experiences

Data Engineer/Jr Technical Instructor, Jarvis (Nov 2025 - Present): Engineered and maintained end-to-end data solutions across software development and data engineering initiatives, leveraging Databricks, PySpark, Delta Live Tables,

and Medallion Architecture to build scalable ETL pipelines and transform CRM and operational data into analytics-ready datasets. Developed backend applications and automation tools using Java, Python, Bash, PostgreSQL, Docker, and Git, while supporting trainee development through the delivery and enhancement of technical training modules, mentorship, and candidate assessments.

Education

McGill University (Sep 2021 - Apr 2025), Bachelor of Science, Statistics and Computer Science

Miscellaneous

- Google Advanced Data Analytics Certificate (July 2025)
- Microsoft Power BI Data Analyst Specialization (August 2025)
- Google Business Intelligence (August 2025)
- IBM Data Engineering Certificate (October 2025)
- Data Engineering with Snowflake Certificate (October 2025)
- Experience with tutoring and swim instructing
- Passion for video editing